

60_wrapper_tmux_bin_valid.sh

Revision: 1 **Last modified:** 2026-06-13T00:00:00Z **Authority:** vasic-digital tmux project **Maintainer:** milosvasic **Scope:** §11.4.18 script companion doc for scripts/tests/60_wrapper_tmux_bin_valid.sh

Overview

Permanent §11.4.135 regression guard for **Issues.md F1** — tmx new -s HelixCode crashed the **whole terminal** on every emulator.

Captured root cause (F1)

The generated/installed wrapper scripts/tmx carried TMUX_BIN=<path> pointing at a **non-existent** binary (a stale prior-checkout location, e.g. /Users/milosvasic/Projects/tmux/tmux/build-darwin/bin/tmux). The operator shell-init runs:

```
exec sh -c 'tmx attach -t NAME 2>/dev/null || exec tmx new -s NAME'
```

tmx new reaches an exec "\$TMUX_BIN" ... in the wrapper (around scripts/tmx.template lines 396 / 430) on the **missing** binary → exec fails ("No such file or directory") → the operator's **login shell** (which had exec'd into the wrapper chain) **dies** → the terminal window closes = "crashes the whole terminal", emulator-independent. v1.0.22 setup.sh regenerated scripts/tmx with a valid TMUX_BIN → fixed.

This test makes that failure mode **impossible to ship silently again**.

What it checks (TDD / anti-bluff, §11.4.43 RED→GREEN)

Layer	Check	Evidence
T1	STATIC always-on guard: the generated scripts/tmx (if present) declares TMUX_BIN=<path> AND that path exists + is executable. SKIP-with-reason when scripts/tmx is absent (not yet generated).	[evidence 60-T1] TMUX_BIN=... exists+executable
T2	RED proof (gives the guard teeth): synthesize a throwaway wrapper copy whose TMUX_BIN is rewritten to a	[evidence 60-T2 ...] RED reproduced ... 'No such file or

Layer	Check	Evidence
	guaranteed-missing path; drive the exact operator path through a real PTY ; assert the captured output names the missing binary with " No such file or directory " at the wrapper's exec "\$TMUX_BIN" sites. Reproduces F1.	directory' naming <missing>
T3	GREEN: with the real valid-TMUX_BIN wrapper, drive the same operator path; assert the session is created (server came up, listed) and the run emits no missing-binary error.	[evidence 60-T3 ...] GREEN: server up ... NO 'No such file or directory'

Each of T2 and T3 runs 3 internal iterations for §11.4.50 deterministic consistency.

Prerequisites

- scripts/tmx generated by scripts/setup.sh (else T1 + T2/T3 SKIP-with-reason per §11.4.3).
- A real tmux binary under tmux/build-darwin|build-linux|build/bin/tmux (or on PATH) for the T3 GREEN server (else T2/T3 SKIP).
- python3 for the PTY operator-path driver (else T2/T3 SKIP).

Usage

```
bash scripts/tests/60_wrapper_tmux_bin_valid.sh
```

Exit 0 = all executed assertions PASS; 1 = a FAIL; 77 = unsupported platform SKIP.

Isolation + cleanup (§11.4.14)

Everything runs under a **short-path** private scratch root /tmp/tmxg.\$\$ with throwaway HOME + TMUX_TMPDIR. The T2 RED uses a throwaway wrapper copy + a guaranteed-missing TMUX_BIN. The T3 GREEN creates and kills **only** its own uniquely-named session/socket (tmxg60_<pid>_<iter>). The trap ... EXIT cleanup removes the scratch root and never runs a broad pkill tmux, never touches the operator's /tmp/tmux-501 or real sessions.

Internal behaviour

The PTY driver (_drive_operator_path) pty.fork()s and execvps sh -c 'exec sh -c \''tmx attach -t "\$1" ... || exec tmx new -s "\$1"\' ...' — byte-for-byte the operator path from scripts/tmx-shell-init.sh.template line 157 — with the chosen wrapper directory placed first on PATH. It captures the controlling PTY's output and the child exit status, returning rc=<code>|out=<captured>. The deterministic F1 discriminator is the captured

"No such file or directory" naming the missing `TMUX_BIN` (present only in T2, never in T3); the child rc varies by branch (bash exec-failure-continue vs sh 127) so it is logged but not asserted on.

Related

- `scripts/tmx.template` — exec `"$TMUX_BIN"` sites (~396 attach / ~430 new).
- `scripts/tmx-shell-init.sh.template` — the operator exec `sh -c ...` path.
- `scripts/verify.sh` — CM-TMX-WRAPPER-TMUXBIN-VALID Layer-1 source gate (refuses to bless a present-but-broken wrapper).
- `scripts/tests/meta_test_false_positive_proof.sh` — M-WRAPPER-TMUXBIN paired §1.1 mutation (points `TMUX_BIN` at a missing path → test 60 FAILs → mutation CAUGHT).
- `scripts/tests/58_operator_path_select_copy_ls.sh` — sibling operator-path proof.
- `Issues.md` F1.

Sources verified 2026-06-13

- In-repo captured forensic evidence: the wrapper exec sites at `scripts/tmx.template` lines 396 / 430 and the operator path at `scripts/tmx-shell-init.sh.template` line 157. NO external solution found — original work.

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